

Companion to Bird & de Moor, Algebra of Programming

§1. Programs

§1.1. Datatypes

§1.2. Natural numbers

§1.3. Lists

§1.4. Trees

§1.5. Inverses

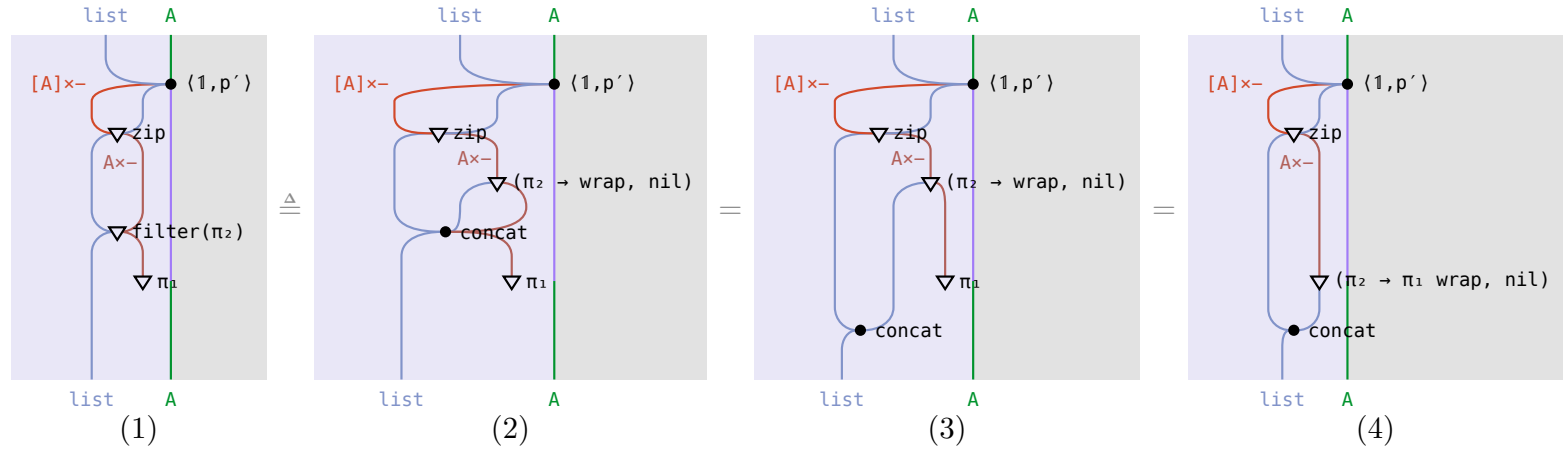
§1.6. Polymorphic functions

§1.7. Pointwise and point-free

$$\langle 1, p' \rangle \text{ zip filter}(\pi_2) \pi_1' = \text{filter}(p) \quad (1.7a)$$

pairing each element with its answer under p ($\langle 1, p' \rangle \text{ zip}$), keeping the pairs answered **true** ($\text{filter}(\pi_2)$) and dropping the answers (π_1') is $\text{filter}(p)$

the starting composite



1	$\langle 1, p' \rangle \text{ zip filter}(\pi_2) \pi_1'$	definition of filter: $\text{filter}(p) \triangleq (p \rightarrow \text{wrap}, \text{nil})'$ concat
2	$\triangleq \langle 1, p' \rangle \text{ zip } (\pi_2 \rightarrow \text{wrap}, \text{nil})' \text{ concat } \pi_1'$	(1.6): $\text{concat } f' = f' \text{ concat}$ (1.8) backwards: $f' g' = (fg)'$
3	$= \langle 1, p' \rangle \text{ zip } (\pi_2 \rightarrow \text{wrap}, \text{nil}) \pi_1'' \text{ concat}$	(1.11): $(p \rightarrow f, g)h = (p \rightarrow fh, gh)$; (1.5): $\text{wrap } f' = f$ wrap; (1.4): $\text{nil } f' = \text{nil}$
4	$= \langle 1, p' \rangle \text{ zip } (\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})' \text{ concat}$	(1.9) backwards: $1 = 1'$
5	$= \langle 1', p' \rangle \text{ zip } (\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})' \text{ concat}$	(1.7): $\langle f', g' \rangle \text{ zip} = \langle f, g \rangle'$ (1.8) backwards: $f' g' = (fg)'$
6	$= \langle 1, p \rangle (\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})' \text{ concat}$	(1.10): $h(p \rightarrow f, g) = (hp \rightarrow hf, hg)$; (1.1): $\langle f, g \rangle \pi_1 = f$; (1.3): $f \text{ nil} = \text{nil}$ (1.12): $1 f = f$
7	$= (p \rightarrow \text{wrap}, \text{nil})' \text{ concat}$	definition of filter
8	$= \text{filter}(p)$	